

Note on UOB Project

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Abstract

Here, we include all the process on this topic. We will start by putting the Primary example, and step by step go on.

Contents

1 Introduction

Finite-dimensional C^* -algebras

Definition 1.1 ([?, §3.2]). A *finite-dimensional C^* -algebra* A is $*$ -isomorphic to a finite direct sum of full matrix algebras

$$A \cong \bigoplus_{i=1}^s M_{n_i}(\mathbb{C}), \quad (n_i \geq 1).$$

Structure theorem. Every finite-dimensional C^* -algebra is semisimple; its simple components are the matrix blocks above and its **minimal central projections**

$$p_i = (0, \dots, 0, I_{n_i}, 0, \dots, 0)$$

sum to 1_A .

Dimension vector. Write $\tilde{n} = (n_1, \dots, n_s)^\top$, a *column* vector. For each i , the irreducible left A -module $V_i = \mathbb{C}^{n_i}$ satisfies $\dim_{\mathbb{C}} V_i = n_i$.

Definition 1.2. Let $B = \bigoplus_{j=1}^r M_{m_j}(\mathbb{C})$ be another finite-dimensional C^* -algebra with dimension vector

$$\tilde{m} = (m_1, \dots, m_r)^\top.$$

A *unital inclusion* $B \subseteq A$ is specified by the *inclusion matrix*

$$H = [h_{ij}] \in M_{s \times r}(\mathbb{N}_0)$$

whose entry h_{ij} is the multiplicity of the simple B -module $W_j = \mathbb{C}^{m_j}$ inside $V_i|_B$. Equivalently

$$H \tilde{m} = \tilde{n}. \tag{1}$$

Conditional expectations and traces

Definition 1.3 (Conditional expectation). Let A be a unital C^* -algebra and $B \subseteq A$ a unital (i.e. $1_B = 1_A$) C^* -subalgebra. A *conditional expectation* (of A onto B) is a linear map

$$E : A \longrightarrow B$$

satisfying the following four properties:

1. **Completely positive:** for every $n \in \mathbb{N}$ the amplified map

$$E^{(n)} : M_n(A) \rightarrow M_n(B), [x_{ij}] \mapsto [E(x_{ij})]$$

sends positive matrices to positive matrices.

2. **Unital (unit-preserving):** $E(1_A) = 1_B$.

3. **B -bimodular:**

$$E(b_1 x b_2) = b_1 E(x) b_2 \quad \text{for all } b_1, b_2 \in B, x \in A.$$

4. **Idempotent (projection):** $E^2 = E$ (equivalently $E|_B = \text{id}_B$).

Canonical (Markov) trace. Write tr_{n_i} for the normalised matrix trace on $M_{n_i}(\mathbb{C})$. The standard trace on A is

$$\tau_A((x_1, \dots, x_s)) = \sum_{i=1}^s \frac{n_i}{d} \text{tr}_{n_i}(x_i), \quad d = \sum_{i=1}^s n_i^2.$$

Jones–Sunder call τ_A the *Markov trace* because $\tau_A \circ E = \tau_A$ for the unique trace-preserving expectation E onto a unital subalgebra B . Its *trace vector* is precisely $\tilde{n}$.

Theorem 1.4 (Jones–Watatani conditional expectation). *Let*

$$A = \bigoplus_{i=1}^s M_{n_i}(\mathbb{C}), \quad B = \bigoplus_{j=1}^r M_{m_j}(\mathbb{C}) \subseteq A,$$

and let

$$\tau_A((x_1, \dots, x_s)) = \frac{1}{d} \sum_{i=1}^s n_i \text{tr}_{n_i}(x_i), \quad d = \sum_{i=1}^s n_i^2,$$

be the **Markov trace**. There exists a unique trace-preserving conditional expectation $E_\tau : A \rightarrow B$. On the i -th block $x_i \in M_{n_i}(\mathbb{C})$ it is given by

$$E_\tau(x_i) = \bigoplus_{j=1}^r \frac{m_j}{n_i} \sum_{\ell=1}^{h_{ij}} (\theta_{ij}^\ell)^{-1} (e_{ij}^\ell x_i e_{ij}^\ell) \quad (1 \leq i \leq s), \quad (2)$$

where $e_{ij}^\ell := \theta_{ij}^\ell(I_{m_j})$ is the supporting projection of the ℓ -th copy of M_{m_j} inside M_{n_i} .

Notation at a glance.

- $H = [h_{ij}]$ is the inclusion matrix; h_{ij} counts how many times M_{m_j} appears inside M_{n_i} .
- $\theta_{ij}^\ell : M_{m_j}(\mathbb{C}) \hookrightarrow M_{n_i}(\mathbb{C})$ ($\ell = 1, \dots, h_{ij}$) is the ℓ -th block-diagonal embedding.
- $e_{ij}^\ell = \theta_{ij}^\ell(I_{m_j})$ is the corresponding *corner projection*. The compression $e_{ij}^\ell x_i e_{ij}^\ell$ lies in the corner algebra $e_{ij}^\ell M_{n_i} e_{ij}^\ell$, which the inverse map $(\theta_{ij}^\ell)^{-1}$ identifies with M_{m_j} .
- The scalar factor m_j/n_i is chosen so that $\tau_A(x) = \tau_A(E_\tau(x))$ for all $x \in A$.

Proof of Theorem ??. We split the argument into six transparent steps.

Step 1: Block notation and embeddings. Fix i . Inside the simple block $M_{n_i}(\mathbb{C})$ the inclusion $B \subseteq A$ picks out h_{ij} mutually orthogonal corners

$$e_{ij}^\ell = \theta_{ij}^\ell(I_{m_j}) \quad (1 \leq j \leq r, 1 \leq \ell \leq h_{ij}), \quad \sum_{j,\ell} e_{ij}^\ell = I_{n_i}.$$

Each θ_{ij}^ℓ is a unital $*$ -embedding

$$\theta_{ij}^\ell : M_{m_j} \hookrightarrow M_{n_i}$$

and $e_{ij}^\ell M_{n_i} e_{ij}^\ell$ inherits the matrix-algebra structure of M_{m_j} through $(\theta_{ij}^\ell)^{-1}$. Explicitly, we may pick matrix units $\{u_{ab}^{(\ell)}\}_{a,b=1}^{m_j} \subset M_{n_i}$ so that $\theta_{ij}^\ell(E_{ab}) = u_{ab}^{(\ell)}$ and $e_{ij}^\ell = u_{11}^{(\ell)} + \dots + u_{m_j m_j}^{(\ell)}$.

Step 2: Defining E_τ . Given $x = (x_1, \dots, x_s) \in A$ set

$$E_\tau(x) = (E_\tau(x_1), \dots, E_\tau(x_s))$$

with $E_\tau(x_i)$ given by formula (??). Because direct sums are orthogonal, it suffices to verify all properties block-wise, i.e. for a *single* i and $x_i \in M_{n_i}$. **Step 3: Well-defined and B -bimodular.** For fixed i , each term

$$(\theta_{ij}^\ell)^{-1}(e_{ij}^\ell x_i e_{ij}^\ell)$$

lies in $M_{m_j}(\mathbb{C})$, hence the orthogonal direct sum over j is in B . Further, for $b_1, b_2 \in B$ their i -th block commutes with every e_{ij}^ℓ (they act *diagonally* on the same minimal central projections). Consequently

$$E_\tau(b_1 x b_2) = b_1 E_\tau(x) b_2, \quad x \in A,$$

so E_τ is B -bimodular.

Step 4: Complete positivity. Each corner map

$$C_{ij}^\ell : x_i \mapsto e_{ij}^\ell x_i e_{ij}^\ell$$

is completely positive (it is a conjugation by a projection). So is its composition with the $*$ -isomorphism $(\theta_{ij}^\ell)^{-1}$. A finite direct sum and a positive scalar multiple of completely positive maps is completely positive. Hence E_τ is *completely* positive.

Step 5: Trace preservation. Compute on a *matrix unit* $u_{ab}^{(\ell)}$ (where $a, b \leq m_j$):

$$E_\tau(u_{ab}^{(\ell)}) = \frac{m_j}{n_i} (\theta_{ij}^\ell)^{-1}(u_{aa}^{(\ell)} u_{ab}^{(\ell)} u_{bb}^{(\ell)}) = \frac{m_j}{n_i} E_{ab} \in M_{m_j}.$$

Taking the trace,

$$\tau_A(u_{ab}^{(\ell)}) = \frac{n_i}{d} \delta_{ab},$$

whereas $\tau_A(E_\tau(u_{ab}^{(\ell)})) = \frac{m_j}{d} \delta_{ab}$. Because $u_{ab}^{(\ell)}$ sits inside the j -th block of B exactly m_j/n_i times, the two numbers coincide. By linearity and faithfulness of τ_A this equality extends from matrix units to all of A :

$$\tau_A \circ E_\tau = \tau_A.$$

Step 6: Uniqueness. Suppose $F : A \rightarrow B$ is another conditional expectation with $\tau_A \circ F = \tau_A$. Let $\{e_\mu\}$ be *all* matrix units in every block M_{n_i} ; these form a basis of A as a vector space. For each μ ,

$$\tau_A((F - E_\tau)(e_\mu)^*(F - E_\tau)(e_\mu)) = 0$$

so $(F - E_\tau)(e_\mu) = 0$ by faithfulness. Hence $F = E_\tau$ on a basis, thus on all of A . $\square$

1.1 The Jones basic construction for $B \subseteq A$

Standing data. Throughout we keep the notation of Theorem ??:

$$A = \bigoplus_{i=1}^s M_{n_i}(\mathbb{C}), \quad B = \bigoplus_{j=1}^r M_{m_j}(\mathbb{C}) \subseteq A, \quad H = [h_{ij}]_{i,j} \in M_{s \times r}(\mathbb{N}_0),$$

and we work in the GNS Hilbert space $L^2(A) := L^2(A, \tau_A)$ where the inner product is $\langle x, y \rangle = \tau_A(y^*x)$. Left multiplication embeds A faithfully into $\mathcal{B}(L^2(A))$.

The Jones projection. Let $E_\tau : A \rightarrow B$ be the trace-preserving conditional expectation from Theorem ?. Define

$$e_B : L^2(A) \longrightarrow L^2(A), \quad e_B(\xi) = E_\tau(\xi) \quad (\xi \in A \subseteq L^2(A))$$

Because E_τ is completely positive and unital, e_B extends by continuity to a bounded operator on all of $L^2(A)$.

Proposition 1.5 (Basic facts about e_B). *The operator e_B satisfies $e_B = e_B^* = e_B^2$ and, for every $x \in A$,*

$$e_B x e_B = E_\tau(x) e_B \quad (\text{Jones relations}).$$

Proof. (i) Projection. For $\xi \in A$ we have $e_B^2(\xi) = E_\tau(E_\tau(\xi)) = E_\tau(\xi) = e_B(\xi)$ because E_τ is idempotent; adjointness is obvious from $\langle e_B \xi, \eta \rangle = \langle \xi, e_B \eta \rangle$.

(ii) Jones relations. Take $x \in A$ and $\xi \in A$. Then

$$e_B x e_B(\xi) = e_B(x E_\tau(\xi)) = E_\tau(x E_\tau(\xi)) = E_\tau(x) E_\tau(\xi) = E_\tau(x) e_B(\xi),$$

where we used the B -bimodule property of E_τ . □

The first basic construction. Set

$$A_1 := C^*(A, e_B) = \text{span}_{C^*} \{a_1 e_B a_2 : a_1, a_2 \in A\}.$$

By Proposition ?? we obtain a new tower of unital inclusions

$$B \subseteq A \subseteq A_1$$

Since A is finite-dimensional, the C^* -algebra A_1 is finite-dimensional as well. We now describe its structure block-by-block.

Theorem 1.6 (Structure of A_1). *Relative to the pair (A, B) the inclusion matrix of $A \subseteq A_1$ is the Gram matrix $H^\top H$. Concretely,*

$$A_1 \cong \bigoplus_{i=1}^s M_{n_i}(\mathbb{C}) \oplus \bigoplus_{j=1}^r h_{ij}^2 M_{m_j}(\mathbb{C}).$$

Idea of proof. For each pair (i, j) the corner $e_{ij}^\ell M_{n_i} e_{ij}^\ell$ is $*$ -isomorphic to M_{m_j} . Products $a_1 e_B a_2$ with $a_k \in M_{n_i}$ are supported on those corners, and the Jones relations force exactly h_{ij}^2 copies of M_{m_j} to appear. A direct matrix computation shows that the multiplicity of the i -th simple block of A inside A_1 is $\sum_j h_{ij}^2$, i.e. the (i, i) -entry of $H^\top H$. □

Index of the inclusion. Set

$$[A : B] := \tau_A(e_B)^{-1}.$$

Because $\tau_A(e_B) = 1/[A : B]$ is simply the norm-square of the inclusion matrix, we get

$$[A : B] = \|H\|_2^2 = \sum_{i=1}^s \sum_{j=1}^r h_{ij}^2.$$

This positive integer is called the **Jones index** of the finite-dimensional inclusion $B \subseteq A$.

Remark 1.7 (Key properties at a glance).

- E_τ is the unique τ_A -preserving conditional expectation from A onto B .
- e_B encodes both E_τ and the inclusion $B \subseteq A$.
- The sequence $B \subseteq A \subseteq A_1 \subseteq A_2 \subseteq \cdots$, obtained by iterating the construction, carries complete information about the inclusion; its abstract combinatorial shadow is the *Bratteli diagram* whose first incidence matrix is H .

2 Need things from Bachi and Bhat paper

Proposition 2.1 (Necessary conditions for a unitary orthonormal basis).

Let $B = \bigoplus_{j=1}^r M_{m_j}(\mathbb{C}) \subseteq A = \bigoplus_{i=1}^s M_{n_i}(\mathbb{C})$ be a unital inclusion with inclusion matrix $H = [h_{ij}]$, dimension vectors

$$\widetilde{\mathbf{m}} = (m_1, \dots, m_r)^\top, \quad \widetilde{\mathbf{n}} = (n_1, \dots, n_s)^\top$$

and conditional expectation $E: A \rightarrow B$. If $(B \subseteq A, E)$ admits a **unitary orthonormal basis** of size d (in the sense of Pimsner–Popa), then

(1) **Spectral condition**

$$H^\top \widetilde{\mathbf{n}} = d \widetilde{\mathbf{m}}, \quad H^\top H \widetilde{\mathbf{m}} = d \widetilde{\mathbf{m}}, \quad HH^\top \widetilde{\mathbf{n}} = d \widetilde{\mathbf{n}}.$$

(2) **Trace (Markov) condition** E preserves the Markov trace

$$\tau_A((x_i)) = \frac{1}{\sum_i n_i^2} \sum_i n_i \operatorname{Tr}(x_i),$$

and E is the unique expectation with this property.

(3) **Integrality of the norm**

$$d = \|H\|_2^2 = \sum_{i=1}^s \sum_{j=1}^r h_{ij}^2 \in \mathbb{N}.$$

(4) **Quadratic dimension identity**

$$\sum_{i=1}^s n_i^2 = d \sum_{j=1}^r m_j^2.$$

3 Finite-Dimensional setup & main results (Bakshi–Bhat)

Standing data

- $B = \bigoplus_{j=1}^r M_{m_j}(\mathbb{C})$, $A = \bigoplus_{i=1}^s M_{n_i}(\mathbb{C})$ (finite-dimensional C^* - or von Neumann algebras).
- $H = [h_{ij}] \in M_{s \times r}(\mathbb{N}_0)$ (inclusion matrix), $H \widetilde{\mathbf{m}} = \widetilde{\mathbf{n}}$ where $\widetilde{\mathbf{m}} = (m_j)^\top$, $\widetilde{\mathbf{n}} = (n_i)^\top$.
- $E: A \rightarrow B$ a faithful conditional expectation.
- Markov trace on A

$$\tau_A((x_i)) = \frac{1}{\sum_i n_i^2} \sum_{i=1}^s n_i \operatorname{Tr}(x_i), \quad \text{trace vector} = \widetilde{\mathbf{n}}.$$

- A **unitary orthonormal basis** is a family $\{W_\alpha\}_{\alpha=1}^d \subset A$ of unitaries with $E(W_\alpha^* W_\beta) = \delta_{\alpha\beta} 1_B$ and $\sum_\alpha W_\alpha E(W_\alpha^* X) = X$ ($X \in A$).

Core theorem (necessary conditions)

Theorem 3.1 (Bakshi–Bhat; Thm. A). *If such a basis of size d exists, then*

Constructions proving sufficiency in special cases

Theorem 3.2 (Explicit bases).

- (a) **Abelian corner (Theorem B).** *If B is abelian and conditions in Thm. ?? hold, a unitary basis exists (quasi-circulant construction $\{V^t U^t\}$).*
- (b) **Simple corner (Theorem C).**
 - *If $B = M_m(\mathbb{C})$ and E preserves the Markov trace, then $(B \subseteq A, E)$ has a unitary basis.*
 - *If $A = M_n(\mathbb{C})$ and E preserves the normalized trace, then $(B \subseteq A, E)$ has a unitary basis.*
- (c) **Weyl-type families.** *When each non-abelian block of A is $M_n(\mathbb{C})$ and B is abelian, the authors build “multi-Weyl” bases extending the classical $(V^j U^k)_{j,k}$ scheme.*

3.1 Abelian corner: existence of a unitary orthonormal basis

We prove part (a) of Theorem ?. Throughout this subsection

$$B = \mathbb{C}^{m_1} \oplus \cdots \oplus \mathbb{C}^{m_r} \quad (\text{finite dimensional } \mathbf{abelian} \text{ } C^*\text{-algebra}),$$

$$A = M_{n_1}(\mathbb{C}) \oplus \cdots \oplus M_{n_s}(\mathbb{C}) \quad (\text{finite dimensional multi-matrix algebra}),$$

and $E : A \rightarrow B$ is the given trace-preserving conditional expectation. Let

$$H = [h_{ij}]_{1 \leq i \leq s, 1 \leq j \leq r} \in M_{s \times r}(\mathbb{N})$$

be the inclusion matrix (notation from [?]). The hypotheses of Theorem ?? say precisely that

$$H^\top H \widetilde{\mathbf{m}} = d \widetilde{\mathbf{m}}, \quad \widetilde{\mathbf{m}} = \begin{bmatrix} m_1 \\ \vdots \\ m_r \end{bmatrix}, \quad d := \|H\|_2^2 \in \mathbb{N},$$

and that E preserves the Markov / normalised trace.

Step 1: reduce to a single matrix block. Because B is abelian, each summand $M_{n_i}(\mathbb{C})$ of A admits the same inclusion pattern. It is therefore enough to build a unitary orthonormal basis (UOB) in the simple corner

$$M_n(\mathbb{C}) \supseteq B_0 := \text{diag}(\mathbb{C}, \dots, \mathbb{C}) \subset M_n(\mathbb{C}), \quad n := n_1 = \cdots = n_s,$$

and then replicate it block-wise. From now on we write $A = M_n(\mathbb{C})$ and $B = B_0$.

Step 2: pick two special unitaries U, V . Set $d = r$ (number of minimal projections in B). Let $\omega := e^{2\pi i/d}$. Define

$$U := \text{diag}(1, \omega, \omega^2, \dots, \omega^{d-1}) \in M_d(\mathbb{C}), \quad V := (v_{kl})_{k,l=0}^{d-1}, \quad v_{kl} := \frac{1}{\sqrt{d}} \omega^{kl} \in M_d(\mathbb{C}).$$

- U is diagonal and unitary;
- V is the $(d \times d)$ discrete Fourier matrix, hence unitary;
- $VU = \omega UV$ (a one-dimensional Weyl relation).

Step 3: define the “quasi-circulant” family. For each $t = 0, \dots, d-1$ put

$$W_t := V^t U^t \in M_d(\mathbb{C}).$$

Because $UV = \omega^{-1}VU$ and $\omega^d = 1$, the set $\{W_t : 0 \leq t < d\}$ is closed under multiplication: $W_s W_t = \omega^{st} W_{s+t}$.

Step 4: orthonormality with respect to E . Identify B with the diagonal subalgebra of $A = M_d(\mathbb{C})$. The trace-preserving expectation is $E(X) = \frac{1}{d} \text{Tr}(X) I_d$ (Tr = ordinary matrix-trace). For $s, t \in \{0, \dots, d-1\}$,

$$E(W_s^* W_t) = \frac{1}{d} \text{Tr}(W_{t-s}) I_d = \begin{cases} I_d & \text{if } s = t, \\ 0 & \text{if } s \neq t, \end{cases}$$

because $\text{Tr}(W_k) = 0$ whenever $k \not\equiv 0 \pmod{d}$. Hence the W_t 's are orthonormal over B .

Step 5: unitarity and spanning. Each W_t is a product of two unitaries, hence unitary. Since $M_d(\mathbb{C})$ has dimension d^2 over $\mathbb{C}$ and $\{W_t b : 0 \leq t < d, b \in \text{diag}(\mathbb{C}^d)\}$ consists of $d \cdot d = d^2$ linearly independent vectors, the family $\{W_t\}_{t=0}^{d-1}$ spans A as a right B -module.

Step 6: conclusion for the abelian corner. Thus $\{W_t\}_{t=0}^{d-1}$ is a unitary orthonormal basis of $A = M_n(\mathbb{C})$ over its abelian subalgebra B . Re-instating the direct-sum decomposition of A , we obtain a UOB for the original inclusion $B \subseteq A$. $\square$

3.2 Simple corner: the case $B = M_m(\mathbb{C})$ & E Markov

We turn to part (b) of Theorem ?? . First assume

$$B = M_m(\mathbb{C}), \quad A = \bigoplus_{i=1}^s M_{n_i}(\mathbb{C}), \quad E: A \twoheadrightarrow B$$

is the unique B -bimodule projection preserving the Markov trace on each block of A . We will produce an explicit UOB.

Step 1: Tensor-product decomposition. Since B is simple, as a right- B -module one checks

$$A \cong M_m(\mathbb{C}) \otimes \mathbb{C}^N, \quad N := \sum_{i=1}^s n_i^2.$$

Under this identification:

$$B = M_m(\mathbb{C}) \otimes 1_N, \quad A = M_m(\mathbb{C}) \otimes M_N(\mathbb{C}),$$

and the Markov-trace-preserving E becomes

$$E(x \otimes y) = \frac{1}{N} \text{Tr}_N(y) (x \otimes I_N).$$

Step 2: Unitary basis in the “matrix” factor. By the abelian–corner case (see §??), there exists a unitary orthonormal basis $\{W_t\}_{t=0}^{N-1} \subset M_N(\mathbb{C})$ for the inclusion $\mathbb{C}^N \subset M_N(\mathbb{C})$ with respect to the normalized trace on M_N . Equivalently

$$E_N(W_s^* W_t) = \delta_{st} I_N, \quad W_t \in M_N(\mathbb{C}),$$

where $E_N(y) = \frac{1}{N} \text{Tr}_N(y) I_N$.

Step 3: Lift to a basis of A over B . Define

$$U_t := I_m \otimes W_t \in M_m(\mathbb{C}) \otimes M_N(\mathbb{C}) = A, \quad t = 0, \dots, N-1.$$

Then each U_t is unitary in A , and

$$E(U_s^* U_t) = E(I_m \otimes W_s^* W_t) = I_m \otimes E_N(W_s^* W_t) = \delta_{st} (I_m \otimes I_N),$$

so $\{U_t\}$ is orthonormal over B . Finally, since $\{W_t\}$ spans M_N as a right $\mathbb{C}^N$ –module, $\{U_t\}$ spans A as a right B –module, giving the desired UOB.

3.3 Dual case: $A = M_n(\mathbb{C})$ & E normalized trace

Now assume

$$A = M_n(\mathbb{C}), \quad B \subseteq A \text{ arbitrary multi-matrix subalgebra}, \quad E: A \rightarrow B$$

preserves the normalized matrix-trace τ_n . Exactly the same tensor–product argument applies once one observes

$$A \cong \mathbb{C}^M \otimes M_n(\mathbb{C}), \quad B = \mathbb{C}^M \otimes I_n,$$

and one invokes the $B = M_m(\mathbb{C})$ case on the first factor. Explicitly:

$$A = (\mathbb{C}^M) \otimes M_n(\mathbb{C}), \quad B = (\mathbb{C}^M) \otimes I_n, \quad E(f \otimes x) = (E_M(f)) \otimes x,$$

where $E_M: \mathbb{C}^M \rightarrow \mathbb{C}^M$ is the normalized-trace expectation. Pulling back a UOB from $\mathbb{C}^M \subset M_M(\mathbb{C})$ completes the proof. $\square$

3.4 Weyl-type families when B is abelian

Assume

$$A = \bigoplus_{i=1}^s M_{n_i}(\mathbb{C}), \quad B \cong \mathbb{C}^s \quad (\text{so } B \text{ is generated by the minimal central projections } p_i),$$

and that $E: A \rightarrow B$ is the trace-preserving expectation coming from the standard trace on each matrix summand. We will exhibit a unitary basis

$$\{W_{i;\alpha,\beta} : 1 \leq i \leq s, 0 \leq \alpha, \beta < n_i\}$$

of A over B .

Step 1: Choose Weyl unitaries on each summand. For each $i = 1, \dots, s$, pick a pair of generating “Weyl” unitaries

$$U_i, V_i \in M_{n_i}(\mathbb{C}),$$

satisfying

$$U_i V_i = \omega_i V_i U_i, \quad U_i^{n_i} = V_i^{n_i} = I_{n_i}, \quad \omega_i = e^{2\pi i/n_i}.$$

Then

$$\{ U_i^\alpha V_i^\beta : 0 \leq \alpha, \beta < n_i \}$$

is a unitary orthonormal basis of M_{n_i} over its centre (with respect to the normalized trace on M_{n_i}).

Step 2: Embed these into A . Under the identification

$$A \cong (p_1 A p_1) \oplus \cdots \oplus (p_s A p_s), \quad p_i A p_i \cong M_{n_i}(\mathbb{C}),$$

define for each triple (i, α, β) the unitary

$$W_{i;\alpha,\beta} = \underbrace{0 \oplus \cdots \oplus 0}_{i-1} \oplus (U_i^\alpha V_i^\beta) \oplus \underbrace{0 \oplus \cdots \oplus 0}_{s-i} \in A.$$

By construction $W_{i;\alpha,\beta}$ acts nontrivially only on the i -th summand and is hence unitary in A .

Step 3: Check orthonormality over B . Since

$$E(x_1 \oplus \cdots \oplus x_s) = \sum_{i=1}^s \frac{1}{n_i} \operatorname{Tr}(x_i) p_i,$$

one computes for any indices $(i, \alpha, \beta), (j, \gamma, \delta)$

$$E(W_{i;\alpha,\beta}^* W_{j;\gamma,\delta}) = \delta_{ij} E_i((U_i^\alpha V_i^\beta)^* (U_i^\gamma V_i^\delta)) p_i = \delta_{ij} \delta_{\alpha\gamma} \delta_{\beta\delta} p_i,$$

where $E_i(y) = \frac{1}{n_i} \operatorname{Tr}(y) I_{n_i}$ is the normalized-trace expectation on the i -th block. In particular

$$E(W_{i;\alpha,\beta}^* W_{j;\gamma,\delta}) = \delta_{i,j} \delta_{\alpha,\gamma} \delta_{\beta,\delta} (p_1 + \cdots + p_s) = \delta_{(i,\alpha,\beta),(j,\gamma,\delta)} 1_B.$$

Hence the collection $\{W_{i;\alpha,\beta}\}$ is orthonormal over B .

Step 4: Spanning property. By counting dimensions one sees there are $\sum_i n_i^2 = \dim A / \dim B$ unitaries above, so orthonormality forces them to span A as a right B -module. Concretely, any $x = x_1 \oplus \cdots \oplus x_s \in A$ can be written block-wise by the Weyl-basis expansion in each $p_i A p_i$, and hence by the same formulas in A itself.

Conclusion. The unitaries

$$\{ W_{i;\alpha,\beta} \} = \{ 0 \oplus \cdots \oplus U_i^\alpha V_i^\beta \oplus \cdots \oplus 0 \}$$

are unitary, orthonormal over B , and span A . This completes the proof of part (c) of Theorem ??.

□

Applications inside the finite-dimensional tower

- Relative entropy. *If a unitary basis exists then $H(A_1 | A) = \ln(\|H\|_2^2)$ where A_1 is the first Jones basic construction.*
- Depth-2 subfactors. *Depth 2 inclusions with abelian relative commutant satisfy the spectral condition automatically, hence admit a unitary basis.*

Author's conjecture (open problem)

Spectral + trace conditions are **sufficient** in full generality.

Current status:

- **Proved** in the special settings of Theorem ??(a)–(c).
- **Open** when both A and B are genuinely multi-matrix (no simple or abelian side) and H satisfies the PF-eigenvector equations but lacks additional regularity.

Remark 3.3 (Perspective). The paper bridges classical Hadamard/Weyl unitary bases and Pimsner–Popa bases in subfactor theory, isolating spectral–theoretic obstructions and giving positive constructions in broad finite-dimensional landscapes.

4 $\mathbb{C} \oplus M_2(\mathbb{C}) \subset M_3(\mathbb{C}) \oplus M_9(\mathbb{C})$: a 2×2 inclusion of index 18

Example 4.1 (Inclusion matrix and basic data).

Inclusion matrix.

$$H = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}, \quad H^T H = \begin{bmatrix} 10 & 4 \\ 4 & 16 \end{bmatrix}.$$

Dimension vectors.

$$\widetilde{m} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \widetilde{n} = H\widetilde{m} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}.$$

Because $H^T H \widetilde{m} = 18 \widetilde{m}$, the Perron–Frobenius eigenvalue is $d = 18$; hence the Jones index satisfies $[A : B] = 18$.

Explicit embedding. Let $B = \mathbb{C} \oplus M_2(\mathbb{C})$ and $A = M_3(\mathbb{C}) \oplus M_9(\mathbb{C})$. For $\beta \in \mathbb{C}$ and $B_2 \in M_2(\mathbb{C})$

$$\iota(\beta \oplus B_2) = \underbrace{\begin{bmatrix} \beta & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \beta \end{bmatrix}}_{M_3(\mathbb{C})} \oplus \underbrace{\begin{bmatrix} \beta & & & \\ & B_2 & & \\ & & B_2 & \\ & & & B_2 \\ & & & & B_2 \end{bmatrix}}_{M_9(\mathbb{C}) \text{ — one } 1 \times 1 \text{ and four } 2 \times 2 \text{ blocks}}.$$

Proposition 4.2 (Markov traces with the ordinary matrix trace). Let

$$B = \mathbb{C} \oplus M_2(\mathbb{C}), \quad A = M_3(\mathbb{C}) \oplus M_9(\mathbb{C}),$$

and keep the embedding $\iota : B \hookrightarrow A$ from §???. With

$$m_1 = 1, \quad m_2 = 2, \quad n_1 = 3, \quad n_2 = 9, \quad d_B = \sum_j m_j^2 = 5, \quad d_A = \sum_i n_i^2 = 90,$$

define the **Markov traces**

$$\tau_B(\beta \oplus B_2) := \frac{1}{d_B} (m_1 \beta + m_2 \operatorname{Tr} B_2) = \frac{1}{5} (\beta + 2 \operatorname{Tr} B_2)$$

$$\tau_A(\alpha \oplus A_9) := \frac{1}{d_A} (n_1 \operatorname{Tr} \alpha + n_2 \operatorname{Tr} A_9) = \frac{1}{90} (3 \operatorname{Tr} \alpha + 9 \operatorname{Tr} A_9)$$

(where Tr is the usual matrix trace, so $\operatorname{Tr}(I_n) = n$). Then

- (i) τ_B and τ_A are faithful traces on B and A ;
- (ii) $\tau_A \circ \iota = \tau_B$ (no extra scale factor);
- (iii) $\tau_A(1) = \tau_B(1) = 1$; hence both are already states.

Proof. ??. Each τ is a positive linear combination of ordinary traces, so traciality and faithfulness are immediate.

??. Take $x = \beta \oplus B_2 \in B$. Under ι it becomes

$$\iota(x) = \underbrace{\begin{bmatrix} \beta & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \beta \end{bmatrix}}_{\alpha \in M_3} \oplus \underbrace{\begin{bmatrix} \beta & & & \\ & B_2 & & \\ & & B_2 & \\ & & & B_2 \\ & & & & B_2 \end{bmatrix}}_{A_9 \in M_9}.$$

Hence $\text{Tr } \alpha = 3\beta$ and $\text{Tr } A_9 = \beta + 4 \text{Tr } B_2$. Substituting in τ_A gives

$$\tau_A(\iota(x)) = \frac{1}{90} (3 \cdot 3\beta + 9(\beta + 4 \text{Tr } B_2)) = \frac{1}{5} (\beta + 2 \text{Tr } B_2) = \tau_B(x).$$

?? Apply each trace to the unit $1_B = (1, I_2)$:

$$\tau_B(1) = \frac{1}{5} (1 + 2 \cdot 2) = 1, \quad \tau_A(1) = \frac{1}{90} (3 \cdot 3 + 9 \cdot 9) = 1.$$

Thus both traces are normalised states, and no rescaling is necessary. $\square$

Remark 4.3 (Dimension-weighted formula). The coefficients $\frac{m_j}{d_B}, \frac{n_i}{d_A}$ come from the general rule $\tau((x_i)) = \sum_i \frac{\dim(V_i)}{\sum_k \dim(V_k)^2} \text{Tr}(x_i)$, so each simple block contributes in proportion to its dimension. With these weights τ_A extends τ_B through the inclusion map and automatically yields the Jones index $[A : B] = \|H\|_2^2 = 18$.

At-a-glance summary (updated).

$$B = \mathbb{C} \oplus M_2(\mathbb{C}) \stackrel{\iota}{\subset} A = M_3(\mathbb{C}) \oplus M_9(\mathbb{C})$$

$$H = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}, \quad [A : B] = 18.$$

Markov traces (states).

$$\tau_B(\beta \oplus B_2) = \frac{1}{5} (\beta + 2 \text{Tr } B_2), \quad \tau_A(\alpha \oplus A_9) = \frac{1}{90} (3 \text{Tr } \alpha + 9 \text{Tr } A_9).$$

Trace-preserving conditional expectation $E : A \rightarrow B$: Let $A_9 = (a_{ij})$ and let $B_2^{(\ell)}$ for $\ell = 1, \dots, 4$ denote the ℓ -th 2×2 block embedded in M_9 .

$$E(\alpha \oplus A_9) = \left[\frac{1}{3} \text{Tr}(\alpha) + \frac{1}{9} a_{11} \right] \oplus \frac{2}{9} \sum_{\ell=1}^4 B_2^{(\ell)}(A_9)$$

where $B_2^{(\ell)}(A_9)$ is the matrix obtained by restricting A_9 to the ℓ -th block.

[Missing figure: Bratteli_{for example}.eps]

Figure 1: Directed Bratteli diagram of the inclusion $\mathbb{C} \oplus M_2(\mathbb{C}) \subset M_3(\mathbb{C}) \oplus M_9(\mathbb{C})$. Black dots on the left mark the two summands of B : $\mathbb{C}$ (top, dimension 1) and $M_2(\mathbb{C})$ (bottom, dimension 2). Black dots on the right mark the summands of A : $M_3(\mathbb{C})$ (top, dimension 3) and $M_9(\mathbb{C})$ (bottom, dimension 9). Arrows point from each B -vertex to each A -vertex, with multiplicities given by the inclusion matrix $H = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}$: three arrows from $\mathbb{C}$ to M_3 , one arrow from $\mathbb{C}$ to M_9 , and four arrows from M_2 to M_9 .

4.1 Path–algebra realisation of the tower

Definition 4.4 (Minimal vs. minimal–central). Let A be a finite–dimensional C^* -algebra.

- A (nonzero) projection $p \in A$ is *minimal* if there is no nonzero projection $q < p$.
- A central projection $z \in Z(A)$ is *minimal central* if no nonzero central $z' < z$.
- Denote $\pi(A) = \{\text{minimal central projections}\}$ and $\text{Min}(A) = \{\text{all minimal projections}\}$.

In the decomposition

$$A \cong \bigoplus_{p \in \pi(A)} M_{d_p}(\mathbb{C}),$$

each p picks out one simple summand, and the ordinary minimal projections are the rank–one matrix units under a unique p .

Definition 4.5 (Inclusion matrix). For the unital inclusion $\mathcal{B} \subseteq A$ with $\mathcal{B} \cong \bigoplus_{p \in \pi(\mathcal{B})} M_{m_p}(\mathbb{C})$ and $A \cong \bigoplus_{q \in \pi(A)} M_{n_q}(\mathbb{C})$, the *inclusion matrix*

$$H = [h_{p,q}]_{p \in \pi(\mathcal{B}), q \in \pi(A)}, \quad h_{p,q} \in \mathbb{N}_0,$$

records how many copies of the $\mathcal{B}$ -block M_{m_p} sit inside the A -block M_{n_q} . Equivalently, $H \widetilde{m} = \widetilde{n}$, where $\widetilde{m} = (m_p)_{p \in \pi(\mathcal{B})}^\top$, $\widetilde{n} = (n_q)_{q \in \pi(A)}^\top$.

Edge sets and path spaces. For each level $k \geq 0$ let

$$\Omega_k = \{\text{edges in the Bratteli diagram of } \mathcal{B} \subseteq A\},$$

and for $-1 \leq \ell < k$ define

$$\Omega[\ell, k] = \{(\alpha_{\ell+1}, \dots, \alpha_k) : \alpha_i \in \Omega_i, r(\alpha_i) = s(\alpha_{i+1})\}, \quad \Omega[\ell, \infty) = \varprojlim_k \Omega[\ell, k].$$

Hilbert spaces. Set

$$H[\ell, k] := \ell^2(\Omega[\ell, k]), \quad H := H[-1, \infty) = \ell^2(\Omega),$$

with orthonormal basis $\{\delta_\alpha\}$.

Definition 4.6 (Path–algebra blocks). For $n \geq -1$, define the block

$$\mathcal{B}_n := \left\{ x \in \mathcal{B}(H) \mid \exists a \in \mathcal{B}(H[n, \infty)) : x(\alpha, \beta) = \delta_{\alpha_{[-1, n-1]}, \beta_{[-1, n-1]}} a(\alpha_{[n, \infty)}, \beta_{[n, \infty)}) \right\}.$$

Then $\mathcal{B}_{-1} \subset \mathcal{B}_0 \subset \mathcal{B}_1 \subset \dots$ is naturally isomorphic (level-by-level) to $\mathcal{B} \subset A_0 \subset A_1 \subset \dots$.

Remark 4.7. A full system of matrix units in $\mathcal{B}_n$ is

$$e_{\lambda, \mu}(\alpha, \beta) = \delta_{\alpha_{[-1, n]}, \lambda} \delta_{\beta_{[-1, n]}, \mu} \delta_{\alpha_{[n+1, \infty)}, \beta_{[n+1, \infty)}},$$

for $\lambda, \mu \in \Omega[-1, n]$ with $r(\lambda) = r(\mu)$.

Markov trace on $\mathcal{B}_n$. Let $t_q^{(n)} > 0$ be the weight on block $q \in \pi(A_n)$, normalized so $\sum_q t_q^{(n)} n_q = 1$. Then for $x \in \mathcal{B}_n$,

$$\tau_{\mathcal{B}_n}(x) = \sum_{\alpha \in \Omega[-1, n]} t_{r(\alpha)}^{(n)} x(\alpha, \alpha)$$

Expectation $E_{\mathcal{B}_k} : \mathcal{B}_n \rightarrow \mathcal{B}_k$ ($k < n$). For $x \in \mathcal{B}_n$ and paths α, β ,

$$(E_{\mathcal{B}_k} x)(\alpha, \beta) = \delta_{\alpha_{[-1, k]}, \beta_{[-1, k]}} \sum_{\substack{\theta \in \Omega[k, n] \\ s(\theta) = r(\alpha_k)}} \frac{t_{r(\theta)}^{(n)}}{t_{r(\alpha_k)}^{(k)}} x(\alpha_k \cdot \theta, \beta_k \cdot \theta).$$

Definition 4.8 (Jones projection). For $n \geq 0$, the Jones projection $e_{n+1} \in \mathcal{B}(H)$ is defined by

$$e_{n+1}(\alpha, \beta) = \delta_{\alpha_{[-1, n-1]}, \beta_{[-1, n-1]}} \delta_{\alpha_{[n+1, \infty)}, \beta_{[n+1, \infty)}} \sqrt{\frac{t_{r(\alpha_n)}^{(n)}}{t_{s(\alpha_n)}^{(n-1)}}} \delta_{\alpha_n, \beta_n}.$$

Remark 4.9. $e_{n+1} \in A'_{n-1} \cap A_{n+1}$, self-adjoint and idempotent, and satisfies

$$e_{n+1} a e_{n+1} = E_{A_{n-1}}(a) e_{n+1}, \quad a \in A_n.$$

4.2 First Jones basic construction

Recall

$$A_0 = \mathcal{B} = \mathbb{C} \oplus M_2(\mathbb{C}), \quad A_1 = A = M_3(\mathbb{C}) \oplus M_9(\mathbb{C}), \quad H = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}.$$

The first basic construction produces

$$A_2 = C^*(A_1, e_B) \cong M_{18}(\mathbb{C}) \oplus M_{36}(\mathbb{C}),$$

because the new inclusion matrix is the transpose $H^T = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$, and hence the dimension vector of A_2 is

$$H^T \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \begin{bmatrix} 18 \\ 36 \end{bmatrix}.$$

In particular, $\dim A_2 = 18^2 + 36^2$ and the Perron–Frobenius eigenvalue of $H^T H$ remains $d = 18$, matching the Jones index $[A : B] = 18$.

Calculation of e_2 in the Jones Tower

Consider the unital inclusion $\mathbb{C} \oplus M_2(\mathbb{C}) \subset M_3(\mathbb{C}) \oplus M_9(\mathbb{C})$, with the tower: - $A_0 = \mathcal{B} = \mathbb{C} \oplus M_2(\mathbb{C})$, - $A_1 = A = M_3(\mathbb{C}) \oplus M_9(\mathbb{C})$, - $A_2 = C^(A_1, e_B) \cong M_{18}(\mathbb{C}) \oplus M_{36}(\mathbb{C})$.*

We compute the Jones projection e_2 for $A_1 \subset A_2$ using the path algebra formulation.

Step 1: Bratteli Diagram and Inclusion Matrices

The inclusion $A_0 \subset A_1$ has matrix:

$$H = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix},$$

with $\pi(A_0) = \{p_1, p_2\}$, $\pi(A_1) = \{q_1, q_2\}$. For $A_1 \subset A_2$, $\pi(A_2) = \{r_1, r_2\}$, and:

$$H^T = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}.$$

Step 2: Path Algebra Formula

For e_2 (corresponding to the inclusion $A_0 \subset A_1$, so $n = 0$ in the formula for e_{n+1}):

$$e_1(\alpha, \beta) = \delta_{\alpha_{[-1, -1]}, \beta_{[-1, -1]}} \delta_{\alpha_{[1, \infty)}, \beta_{[1, \infty)}} \sqrt{\frac{t_{r(\alpha_0)}^{(0)}}{t_{s(\alpha_0)}^{(-1)}}} \delta_{\alpha_0, \beta_0}$$

where for $\alpha, \beta \in \Omega[-1, \infty)$: - $\alpha = (\alpha_0, \alpha_1, \dots)$ where α_i is an edge from level $i - 1$ to i . - $e_1(\alpha, \beta) \neq 0$ only if the paths agree after level 1 and have the same first edge $\alpha_0 = \beta_0$.

Step 3: Weights

The Markov trace weights for the vertices (simple blocks) are proportional to the dimensions of the corresponding simple modules. Let's normalize $t^{(-1)} = 1$ for the unique block at level -1.

- Level 0: $\widetilde{m} = (1, 2)^T$. The trace vector is $t^{(0)} = (1, 2)^T$.
- Level 1: $\widetilde{n} = (3, 9)^T$. The trace vector is $t^{(1)} = (3, 9)^T$.

The coefficients for the square root term are:

- Path from level -1 to p_1 : $\sqrt{\frac{t_{p_1}^{(0)}}{t^{(-1)}}} = \sqrt{\frac{1}{1}} = 1$.
- Path from level -1 to p_2 : $\sqrt{\frac{t_{p_2}^{(0)}}{t^{(-1)}}} = \sqrt{\frac{2}{1}} = \sqrt{2}$.

Step 4: Matrix Form

The trace of the Jones projection must be $\tau_{A_2}(e_2) = 1/[A_1 : A_0] = 1/18$. The Markov trace on A_2 is $\tau_{A_2}(X_1 \oplus X_2) = \frac{1}{1620}(18 \text{Tr}(X_1) + 36 \text{Tr}(X_2)) = \frac{1}{90} \text{Tr}(X_1) + \frac{1}{45} \text{Tr}(X_2)$. Let $e_2 = P_1 \oplus P_2$, where P_1, P_2 are projections. Their ranks must satisfy $\frac{1}{90} \text{rank}(P_1) + \frac{1}{45} \text{rank}(P_2) = \frac{1}{18}$, which simplifies to $\text{rank}(P_1) + 2 \text{rank}(P_2) = 5$. A possible solution for the ranks is $(\text{rank}(P_1), \text{rank}(P_2)) = (3, 1)$. This leads to the representation:

$$e_2 = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}_{18 \times 18} \oplus \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}_{36 \times 36}.$$

References

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